MTH 201 Chapter 3: Infinite Series

3.2 Worksheet

1. Determine whether each of the following series converge or diverge. If they converge, find what value they converge to.

a) $\sum_{n=1}^{\infty} \frac{7}{2^{n+2}}$

b) $\sum_{k=0}^{\infty} 2^{2k} \cdot 3^{1-k}$

c) $\sum_{k=0}^{\infty} \frac{3^k}{e^{k+1}}$

2. Each of following is a geometric series. Determine if this series converges or diverges. If it converges, then find its sum.

a) $3-4+\frac{16}{3}-\frac{64}{9}+\ldots$

b) $4+3+\frac{9}{4}+\frac{27}{16}+\ldots$

c) $10-2+0.4-0.08+\ldots$

d) $2+0.5+0.125+0.03125+\ldots$

3. $\sum_{k=0}^{\infty}\left(\sqrt{k}-\sqrt{k+1}\right)$

4. $\sum_{n=2}^{\infty}\left(\frac{1}{n-1}-\frac{1}{n+1}\right)$

5. $\sum_{n=1}^{\infty}\left(\cos\left(\frac{1}{n^2}\right)-\cos\left(\frac{1}{(n+1)^2}\right)\right)$

6. $\sum_{n=1}^{\infty}\left(e^{1/n}-e^{1/(n+1)}\right)$

7. $\frac{2}{1\cdot 3}+\frac{2}{2\cdot 4}+\frac{2}{3\cdot 5}+\frac{2}{4\cdot 6}+\frac{2}{5\cdot 7}+\frac{2}{6\cdot 8}+\ldots$ (*Hint: Try writing this series in summation notation*)

8. $\sum_{n=1}^{\infty} \frac{n+1}{n^2+n}$

9. $\sum_{n=2}^{\infty} \frac{1}{n-1}$